

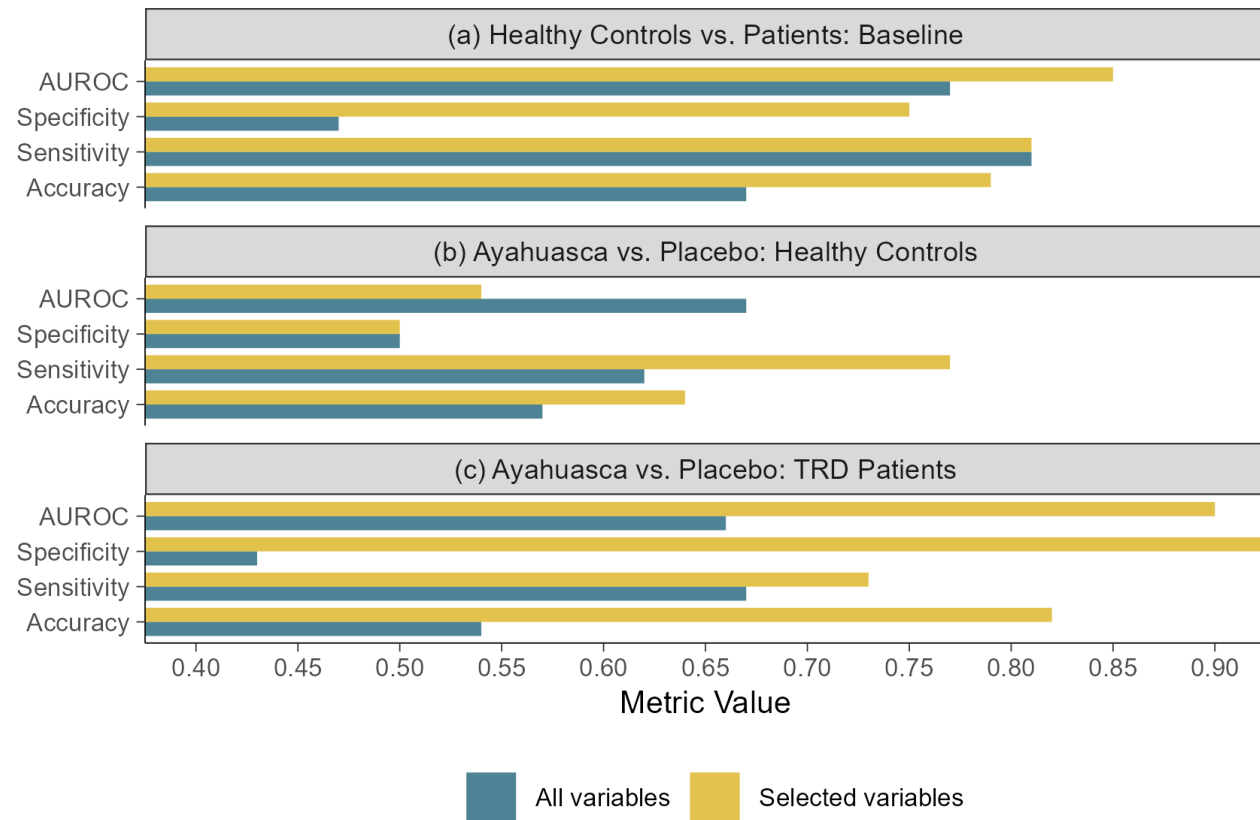
Screening biochemical changes induced by psychedelic ayahuasca in healthy subjects and patients with depression: a machine learning approach

Supplementary Material

Algorithm	Test Fold1	Test Fold2	Test Fold3	Test Fold4	Test Fold 5	Mean Accuracy
Gaussian Naive-Bayes	71.43	78.57	64.29	64.29	71.43	70.00
Random Forest	78.57	64.29	64.29	71.43	57.14	67.14
Logistic Regression	78.57	42.86	57.14	71.43	78.57	65.71
XGBoost	71.43	57.14	57.14	71.43	64.29	64.29
Support Vector Machine	64.29	64.29	57.14	57.14	57.14	60.00
Bernoulli Naive-Bayes	64.29	64.29	57.14	57.14	57.14	60.00
Decision Tree	71.43	57.14	50.00	57.14	57.14	58.57

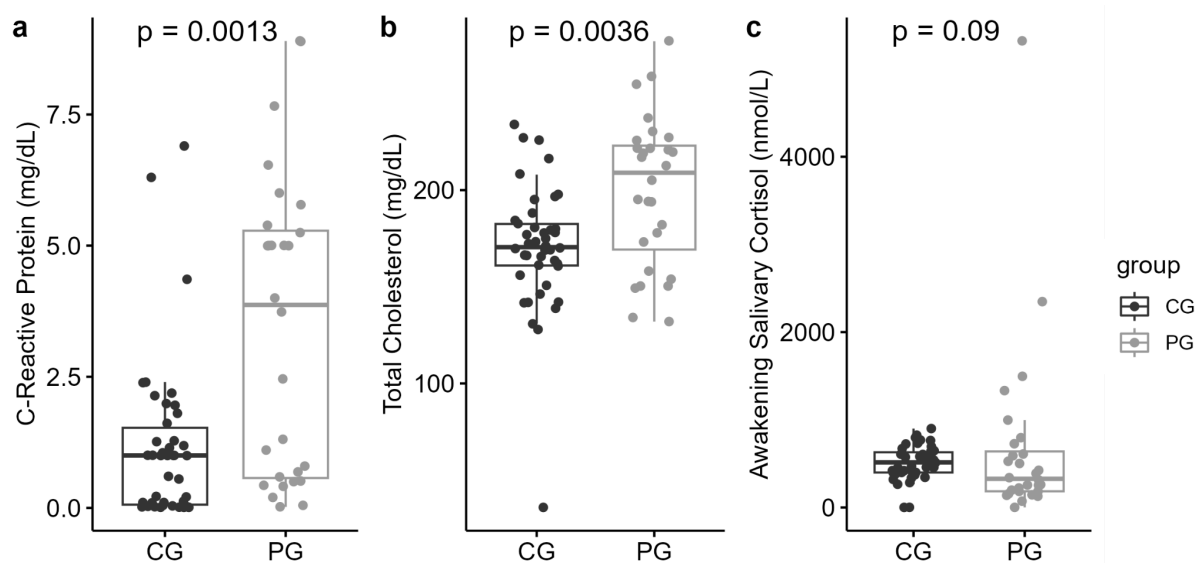
Supplementary Table 1. Comparison of accuracy among different machine learning models.

The table shows the comparison of different machine learning algorithms and their performance in a classification task involving control and patient groups, as well as the comparison of Ayahuasca and Placebo. The Random Forest algorithm demonstrated a competitive accuracy compared to other algorithms though folds. Besides that, other advantages of Random Forest over alternative algorithms is its robustness, ability to handle non-linearity, and feature importance estimation.



Supplementary Figure 1. Comparison of Metrics in RF-5CV with all variables and selected variables at baseline comparison and treatment comparison.

Supplementary Figure 1 showcases a comprehensive comparison of four important machine learning metrics – Accuracy, Sensitivity, Specificity, and Area Under the Receiver Operating Characteristic (AUROC) – derived from Random Forest with 5-fold Cross-Validation (RF-5CV). The model's performance was evaluated using all available variables and a select set of variables, both at baseline (Healthy Controls vs. Patients) and during treatment (Ayahuasca vs. Placebo).



Supplementary Figure 2. Wilcoxon tests of Random Forest 5-CV plus RFE comprising selected variables to compare Controls and TDR patients at baseline. Graphs depict box plots with median and interquartile intervals and dots represent individual measurements. Control groups are in black and patients are represented in gray. a) C-Reactive protein of Control group (CG) and TDR patients group (PG). b) Total Cholesterol of CG and PG. c) Awakening Salivary Cortisol of CG and PG. Above the plots, p-values of Wilcoxon tests are also provided. Please note that the first two comparisons were significant while the Awakening Salivary Cortisol difference is marginally significant ($0.05 < p < 0.1$).

Variable	Group	Median _Before	Median _After	IQR_ Before	IQR_ After	p value	Effect _Size	Confidence Interval
ast	Ayahuasca	19.5	22	9	15.5	0.4139	-0.03	[-0.46, 0.49]
ast	Placebo	19.5	18	7.5	7.75	0.8227	-0.05	[-0.34, 0.34]
bdnf	Ayahuasca	11815.5	10495	4448	3759	0.2166	0.18	[-0.38, 0.93]
bdnf	Placebo	11454	11380.5	6728	5393.2 5	0.8077	-0.15	[-0.38, 0.18]
crp	Ayahuasca	5	2.8	34.375	3.935	0.0056	0.51	[0.21, 1.09]
crp	Placebo	0.95	3.38	4.22	4.515	0.442	-0.17	[-0.55, 0.18]
glucose	Placebo	97	93	14.75	17.75	0.0512	0.09	[0, 0.36]
glucose	Ayahuasca	90.5	95	12.25	12.75	0.2451	-0.22	[-0.78, 0.11]
ldlcholes terol	Placebo	106	110	48.25	57.75	0.2080	-0.08	[-0.28, 0.05]
ldlcholes terol	Ayahuasca	145	146	63	57.75	0.6375	-0.1	[-0.46, 0.18]
lymphoc yte(%)	Placebo	32	35.5	7.5	9.5	0.0214	-0.55	[-1.05, -0.14]
lymphoc yte(%)	Ayahuasca	32.5	30.5	11	11.5	0.4308	0.13	[-0.27, 0.35]
lymphoc yte	Placebo	2126	2401	612	923.25	0.1353	-0.2	[-0.64, 0.07]
lymphoc yte	Ayahuasca	2512.5	2230	916.5	484	0.1937	0.22	[-0.12, 0.77]
rbc	Ayahuasca	4.605	4.54	0.5299	0.565	0.0685	0.25	[0.01, 0.49]
rbc	Placebo	4.68	4.51	0.5525	0.5075	0.3151	0.05	[-0.46, 0.45]

Variable	Group	Median Before	Median After	IQR Before	IQR After	p value	Effect Size	Confidence Interval
lymphocyte (%)	Placebo	32.00	35.50	7.50	9.50	0.0214	0.55	[-1.05, -0.14]
crp	Ayahuasca	5.00	2.80	34.375	3.935	0.0056	0.51	[0.21, 1.09]
plasmatic cortisol	Ayahuasca	147.345	153.67	57.3975	93.7825	0.1040	0.41	[-71.956, 20.552]
rbc	Ayahuasca	4.605	4.54	0.5299	0.565	0.0685	0.25	[0.01, 0.49]
glucose	Ayahuasca	90.50	95.00	12.25	12.75	0.2451	0.22	[-0.78, 0.11]
lymphocyte (absolute)	Ayahuasca	2512.50	2230.00	916.50	484.00	0.1937	0.22	[-0.12, 0.77]
lymphocyte (absolute)	Placebo	2126.00	2401.00	612.00	923.25	0.1353	0.20	[-0.64, 0.07]
bdnf	Ayahuasca	11815.50	10495.00	4448.00	3759.00	0.2166	0.18	[-0.38, 0.93]
crp	Placebo	0.95	3.38	4.22	4.515	0.4420	0.17	[-0.55, 0.18]
plasmatic cortisol	Placebo	154.77	148.78	59.77	82.605	0.5416	0.16	[-69.972, 44.172]
bdnf	Placebo	11454.00	11380.50	6728.00	5393.25	0.8077	0.15	[-0.38, 0.18]
lymphocyte (%)	Ayahuasca	32.50	30.50	11.00	11.50	0.4308	0.13	[-0.27, 0.35]
ldlcholesterol	Ayahuasca	145.00	146.00	63.00	57.75	0.6375	0.10	[-0.46, 0.18]
glucose	Placebo	97.00	93.00	14.75	17.75	0.0512	0.09	[0, 0.36]
ldlcholesterol	Placebo	106.00	110.00	48.25	57.75	0.2080	0.08	[-0.28, 0.05]

ast	Placebo	19.50	18.00	7.50	7.75	0.8227	0.05	[-0.34, 0.34]
rbc	Placebo	4.68	4.51	0.5525	0.5075	0.3151	0.05	[-0.46, 0.45]
ast	Ayahuasca	19.50	22.00	9.00	15.50	0.4139	0.03	[-0.46, 0.49]

Supplementary Table 2. Results of Wilcoxon tests of selected features for the PG (Patients Group) in the study, comparing different biomarkers before and after treatment with Ayahuasca and Placebo. In the patients group, the levels of the selected features were compared "before" and "after" within each treatment (Ayahuasca and Placebo).

Variable	Treatment	Median_Before	Median_After	IQR_Before	IQR_After	p value	Effect_Size	Confidence Interval
ast	Ayahuasca	16.5	15.5	4	3.75	0.2285	0.11	[-0.15, 0.49]
ast	Placebo	17	18	2.75	6	0.3906	-0.34	[-0.93, 0.05]
bdnf	Placebo	10905	9761.5	2617.75	3629.25	0.1044	0.42	[0.08, 1]
bdnf	Ayahuasca	11466	11715	4778.25	3100	0.6556	0.14	[-0.3, 0.42]
crp	Ayahuasca	1	0.37	10.925	12.425	0.0365	0.25	[0.09, 0.65]
crp	Placebo	0.61	0.23	1.62	14.325	0.2440	-0.07	[-0.38, 0.15]
glucose	Ayahuasca	88	89	11.5	7.75	0.6008	0.01	[-0.34, 0.43]
glucose	Placebo	85.5	88	9.25	9	0.6315	0.15	[-0.46, 0.66]
ldlcholesterol	Placebo	100	100	22.5	28.5	0.3336	0.08	[-0.09, 0.23]
ldlcholesterol	Ayahuasca	105	106.5	18.75	13.25	0.5896	-0.11	[-0.69, 0.27]
lymphocyte(%)	Placebo	34	35.5	7.75	13.25	0.7312	-0.02	[-0.41, 0.42]

lymphocyte(%)	Ayahuasca	36	32.5	10.75	7.5	0.7787	-0.02	[-0.23, 0.29]
lymphocyte	Placebo	2226	2325	661.5	594	0.6215	-0.13	[-0.6, 0.4]
lymphocyte	Ayahuasca	2051	2106.5	867.25	480.75	0.8987	0.04	[-0.25, 0.33]
rbc	Ayahuasca	4.915	4.84	0.9125	0.7325	0.0424	0.2	[0.03, 0.41]
rbc	Placebo	4.82	4.76	0.4625	0.7675	0.6011	0.01	[-0.16, 0.16]

Supplementary table 3. Results for the same selected features but for the CG (Control Group) in the study, comparing different biomarkers before and after treatment with Ayahuasca and Placebo.

In the control group of the placebo-controlled experiment, the levels of two biomarkers, C-Reactive Protein (CRP) and Red Blood Cells (RBC), were compared "before" and "after" within each treatment (Ayahuasca and Placebo).

Biomarker	Placebo Median	Ayahuasca Median	Placebo IQR	Ayahuasca IQR	P-value	Effect Size
glucose	-3.0	5.0	6.75	12.0	0.078	0.976
crp	0.06	-0.845	2.407	1.98	0.058	0.448
lymphocyte (%)	5.0	-1.0	9.25	6.5	0.054	1.086
lymphocyte (absolute)	216.0	-125.0	337.5	270.75	0.217	0.781
tgo	0.5	1.5	3.25	10.25	0.504	0.114
bdnf	320.5	-1013.5	2710.0	2689.5	0.502	0.353
rbc	-0.02	-0.12	0.442	0.285	0.326	0.378
ldl cholesterol	1.5	4.0	10.25	23.25	1.000	0.137
plasmatic	12.855	14.615	91.695	72.0175	0.646	-0.195

cortisol						
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Supplementary Table 4. Comparative Analysis of Biomarker deltas between Placebo and Ayahuasca Treatments. Supplementary Table 4 presents the results of a Wilcoxon test comparing delta values for various biomarkers between two treatment groups: Placebo and Ayahuasca. The table provides valuable insights into the effects of Ayahuasca treatment on different biomarkers.

Biomarker	Correlation	p-value
bdnf	-0.607769	0.021131
plasmatic cortisol	0.601486	0.022879
lymphocyte(absolute)	0.441802	0.113738
hcm	-0.422750	0.132075
neutrophils	-0.416497	0.138501
alt	0.338540	0.236425
ldlcholesterol	0.325334	0.256368
lymphocyte	0.285922	0.321700
hematocrit	0.265885	0.358206
ast	0.251709	0.385338
total cholesterol	0.227444	0.434211
glucose	0.225098	0.439094
Monocytes	0.207433	0.476726

(absolute)		
triglycerides	-0.199550	0.493996
ureia	0.182583	0.532130
potassium	0.160853	0.582771
hdl cholesterol	-0.140653	0.631509
sodium	-0.051270	0.861816
auc salivar	0.028235	0.923668

Supplementary Table 5: Pearson correlation coefficients between delta MADRS and selected features by Random forest regression with RFECV for Ayahuasca in TRD Patients

This table provides the Pearson correlation coefficients and corresponding p-values for the Ayahuasca treatment. Each row corresponds to a specific biomarker. The 'Correlation' column lists the correlation coefficients between each biomarker and the 'deltamadr' score. The 'p-value' column provides the corresponding p-values for these correlations.

Biomarker	Correlation	p-value
potassium	0.530275	0.051102
Monocytes (absolute)	0.454722	0.102343
ureia	-0.320289	0.264245
ast	0.312863	0.276100
hematocrit	0.285957	0.321636
alt	0.237997	0.412586
plasmatic cortisol	-0.229171	0.430632
bdnf	0.210871	0.469284

auc salivar	-0.178316	0.541919
lymphocyte	-0.135790	0.643462
neutrophils	-0.135256	0.644781
sodium	-0.125952	0.667888
total cholesterol	0.118608	0.686327
ldlcholesterol	0.088436	0.763693
glucose	0.073165	0.803695
triglycerides	-0.059227	0.840605
lymphocyte (absolute)	-0.057853	0.844261
hcm	0.056468	0.847950
hdl cholesterol	-0.009902	0.973200

Supplementary Table 6: Pearson correlation coefficients between delta MADRS and selected features by Random forest regression with RFECV for Placebo in TRD Patients

This table presents the Pearson correlation coefficients and the corresponding p-values for the Placebo treatment. Each row represents a specific biomarker. The 'Correlation' column provides the correlation coefficients between each biomarker and the 'deltamadr' score. The 'p-value' column lists the corresponding p-values for these correlations.